Mercado, Nicolai

Math 40 THV Quiz 5

1. Determine if the following sets are linearly dependent. If the set is linearly dependent, write one of the vectors as a linear combination of the other vectors. Please use the matrices after item 2.

(a) $\{(1,2,-1,1),(0,1,0,1),(1,4,-1,3),(0,0,1,1),(1,4,3,7)\}$

(1) Can we find $a,b,c,d,e \in \mathbb{R}$ not all 0 such that

$$0 = a(1,2,-1,1) + b(0,1,0,1) + c(1,4,-1,3) + d(0,0,1,1) + e(1,4,3,7)$$

$$= (a+c+e,\ 2a+b+4c+4e,\ -a-c+d+3e,\ a+b+3c+d+7e)$$

This gives the linear system whose coefficient matrix is:

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 2 & 1 & 4 & 0 & 4 \\ -1 & 0 & -1 & 1 & 3 \\ 1 & 1 & 3 & 1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 2 & 0 & 2 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Since there exist free variables, the set is linearly dependent.

(2) Moreover, $a = -c - e$

$b = -2c - 2e$

$d = -4e$

and so if $e = 0$, $c = 1$

$$0 = (-1)(1,2,-1,1) + (-2)(0,1,0,1) + (1,4,-1,3)$$

$$= (-1,-2,1,-1) + (0,-2,0,-2) + (1,4,-1,3)$$

$$= (0,0,0,0)$$

(3) and so at least one of the vectors in the list is a linear combination of the other vectors

(b) $\{x^3 - 2x^2 + 3x + 4, x^2, -x^3 + 3x^2 - 3x - 4, x + 1, x^3 + 4x + 5\}$

(1) Can we find $a,b,c,d,e \in \mathbb{R}$ not all 0 such that

$$0 = a(x^3 - 2x^2 + 3x + 4) + b(x^2) + c(-x^3 + 3x^2 - 3x - 4) + d(x+1) + e(x^3 + 4x + 5)$$

$$= x^3(a - c + e) + x^2(-2a + b + 3c) + x(3a - 3c + d + 4e) + (4a - 4c + d + 5e)$$

This gives the linear system whose coefficient matrix is:

$$\begin{bmatrix} 1 & 0 & -1 & 0 & 1 \\ -2 & 1 & 3 & 0 & 0 \\ 3 & 0 & -3 & 1 & 4 \\ 4 & 0 & -4 & 1 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Since there exist free variables, the set is linearly dependent.

(2) Moreover, $a = -e + c$

$b = -2e - c$

$d = -e$

and so if $c = -1$, $e = 0$

$$0 = (-1)(x^3 - 2x^2 + 3x + 4) + (x^2) + -1(-x^3 + 3x^2 - 3x - 4)$$

$$= -x^3 + 2x^2 - 3x - 4 + 0 + x^2 + 0 + 0 + x^3 - 3x^2 + 3x + 4$$

$$= 0 \checkmark$$

(3) and so at least one of the vectors in the list is a linear combination of the other vectors

(c) $\left\{\begin{bmatrix}1&1\\2&1\end{bmatrix}, \begin{bmatrix}0&1\\0&-1\end{bmatrix}, \begin{bmatrix}1&2\\2&0\end{bmatrix}, \begin{bmatrix}-2&-2\\-3&-1\end{bmatrix}, \begin{bmatrix}4&5\\7&2\end{bmatrix}\right\}$

(1) Can we find $a,b,c,d,e \in \mathbb{R}$ not all 0 such that

$$0 = a\left(\begin{bmatrix}1&1\\2&1\end{bmatrix}\right) + b\left(\begin{bmatrix}0&1\\0&-1\end{bmatrix}\right) + c\left(\begin{bmatrix}1&2\\2&0\end{bmatrix}\right) + d\left(\begin{bmatrix}-2&-2\\-3&-1\end{bmatrix}\right) + e\left(\begin{bmatrix}4&5\\7&2\end{bmatrix}\right)$$

$$= \begin{bmatrix} a+c-2d+4e & a+b+2c-2d+5e \\ 2a+2c-3d+7e & a-b-d-2e \end{bmatrix}$$

This gives the linear system

$$0 = a+c-2d+4e$$
$$0 = a+b+2c-2d+5e$$
$$0 = 2a+2c-3d+7e$$
$$0 = a-b-d-2e$$

the coefficient matrix of the preceeding system

$$\begin{bmatrix} 1 & 0 & 1 & -2 & 4 \\ 1 & 1 & 2 & -2 & 5 \\ 2 & 0 & 2 & -3 & 7 \\ 1 & -1 & 0 & -1 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 0 & 2 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Since there exist free variables, the set is <u>linearly dependent</u>.

(2) Moreover, $a = -2e - c$
$b = -e - c$
$d = e$

and so, if $c = 1$, $e = 0$

$$0 = (-1)\left(\begin{bmatrix}1&1\\2&1\end{bmatrix}\right) + (-1)\left(\begin{bmatrix}0&1\\0&-1\end{bmatrix}\right) + \left(\begin{bmatrix}1&2\\2&0\end{bmatrix}\right)$$

$$= \begin{bmatrix}-1&-1\\-2&-1\end{bmatrix} + \begin{bmatrix}0&-1\\0&1\end{bmatrix} + \begin{bmatrix}1&2\\2&0\end{bmatrix}$$

$$= \begin{bmatrix}0&0\\0&0\end{bmatrix} = 0 \checkmark$$

(3) and so at least one of the vectors in the list is a linear combination of the other vectors

2. Determine if v is an element of the span of S justify your answer. Please use the matrices after item 2.

(a) $v = x^3 + 4x^2 + 3x + 7$, $S = \{x^3 + 2x^2 - x + 1, x^2 + 1, x^3 + 4x^2 - x + 3, x + 1\}$

(1) Can we find an $a,b,c,d \in \mathbb{R}$ such that

$$x^3 + 4x^2 + 3x + 7 = a(x^3 + 2x^2 - x + 1) + b(x^2+1) + c(x^3 + 4x^2 - x + 3) + d(x+1)$$
$$= x^3(a+c) + x^2(2a+b+4c) + x(-a-c+d) + (a+b+3c+d)$$

(2) Equating coefficients, we get a linear system whose augmented matrix is:

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 2 & 1 & 4 & 0 & 4 \\ -1 & 0 & -1 & 1 & 3 \\ 1 & 1 & 3 & 1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 2 & 0 & 2 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(3) Thus, $a = 1 - c$
$b = 2 - 2c$
$d = 4$

In particular,

$$x^3 + 4x^2 + 3x + 7$$
$$= 1(x^3 + 2x^2 - x + 1) + 2(x^2+1) + 0 + 4(x+1)$$
$$= x^3 + 2x^2 - x + 1 + 0 + 2x^2 + 0 + 2 + 0 + 0 + 4x + 4$$
$$= x^3 + 4x^2 + 3x + 7 \checkmark$$

and so v is in the span

(b) $v = \begin{bmatrix} 1 & 0 \\ 4 & 5 \end{bmatrix}$, $S = \left\{ \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} -1 & 3 \\ -3 & -4 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \right\}$

Can we find an $a, b, c, d \in \mathbb{R}$ such that

① $\begin{bmatrix} 1 & 0 \\ 4 & 5 \end{bmatrix} = a\left(\begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}\right) + b\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right) + c\left(\begin{bmatrix} -1 & 3 \\ -3 & -4 \end{bmatrix}\right) + d\left(\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}\right)$

This gives the linear system

$$\begin{aligned} 1 &= a - c \\ 0 &= -2a + b + 3c \\ 4 &= 3a - 3c + d \\ 5 &= 4a - 4c + d \end{aligned}$$

the augmented matrix of the ② preceeding system:

$$\begin{bmatrix} 1 & 0 & -1 & 0 & 1 \\ -2 & 1 & 3 & 0 & 0 \\ 3 & 0 & -3 & 1 & 4 \\ 4 & 0 & -4 & 1 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

③ Thus, $a = 1 + c$
$b = 2 - c$
$d = 1$

In particular,

$$\begin{bmatrix} 1 & 0 \\ 4 & 5 \end{bmatrix} = 1\left(\begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}\right) + 2\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right) + 0 + 1\left(\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}\right)$$

$$= \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

$$\overset{\checkmark}{=} \begin{bmatrix} 1 & 0 \\ 4 & 5 \end{bmatrix}$$

and so v is in the span

(c) $v = (4, 5, 7, 2)$, $S = \{(1, 1, 2, 1), (0, 1, 0, -1), (1, 2, 2, 0), (-2, -2, -3, -1)\}$

① Can we find an $a, b, c, d \in \mathbb{R}$ such that

$$\begin{aligned} (4,5,7,2) &= a(1,1,2,1) + b(0,1,0,-1) + c(1,2,2,0) + d(-2,-2,-3,-1) \\ &= (a + c - 2d,\ a + b + 2c - 2d,\ 2a + 2c - 3d,\ a - b - d) \end{aligned}$$

② Equating coefficients, we get a linear system whose augmented matrix is

$$\begin{bmatrix} 1 & 0 & 1 & -2 & 4 \\ 1 & 1 & 2 & -2 & 5 \\ 2 & 0 & 2 & -3 & 7 \\ 1 & -1 & 0 & -1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 0 & 2 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

③ Thus, $a = 2 - c$
$b = 1 - c$
$d = -1$

In particular,

$$\begin{aligned} (4,5,7,2) &= 2(1,1,2,1) + 1(0,1,0,-1) + 0 + (-1)(-2,-2,-3,-1) \\ &= (2,2,4,2) + (0,1,0,-1) + (2,2,3,1) \\ &\overset{\checkmark}{=} (4,5,7,2) \end{aligned}$$

and so v is in the span